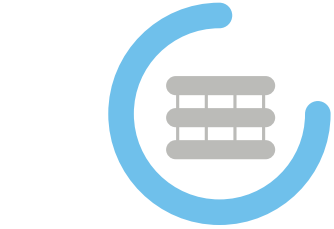

Meeting of the Technical Steering Committee (TSC) Board

Wednesday, January 27th, 2021
11:00am ET

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THE LINUX FOUNDATION

Agenda/Updates

- Announcements, Upcoming talks and deadlines
  - ISC 2021
    - Accepted BoF Sessions of ISC 2020 will be held at ISC 2021!
  - PEARC'21 – virtual conference
    - tutorial submissions due February 9, 2021

- Linaro CI system updates
- Update on co-install gotcha from last time
- ARM compiler update
- CentOS/RHEL discussion
- Cloud working group

Update on Package co-install Gotcha

- recall from last time: issue with %doc files in some of our .spec files preventing co-installation of multiple versions of the same package

```
# rpm -q mfem-gnu9-openmpi4-ohpc
mfem-gnu9-openmpi4-ohpc-4.1-8.1.ohpc.2.0.x86_64

# rpm -ivh mfem-gnu9-openmpi4-ohpc-4.2-1.3.ohpc.2.1.x86_64.rpm
Verifying...                          ################################# [100%]
Preparing...                          ################################# [100%]
file /opt/ohpc/pub/doc/contrib/mfem-gnu9-openmpi4-ohpc/INSTALL from install of mfem-
gnu9-openmpi4-ohpc-4.2-1.3.ohpc.2.1.x86_64 conflicts with file from package mfem-gnu9-
openmpi4-ohpc-4.1-8.1.ohpc.2.0.x86_64
file /opt/ohpc/pub/doc/contrib/mfem-gnu9-openmpi4-ohpc/README from install of mfem-
gnu9-openmpi4-ohpc-4.2-1.3.ohpc.2.1.x86_64 conflicts with file from package mfem-gnu9-
openmpi4-ohpc-4.1-8.1.ohpc.2.0.x86_64
```

- was thinking we might have to update all relevant .spec files for development packages to fix
- fortunately, there looks to be an easier way…

Update on Package co-install Gotcha

- looked at rpm options associated with docdir

```
$ rpmbuild  --showrc | grep docdir
  RPM_DOC_DIR="%{_docdir}"
-13: __docdir_path %{_datadir}/doc:%{_datadir}/man:%{_datadir}/info:%{_datadir}/gtk-
doc/html:%{?_docdir}:%{?_mandir}:%{?_infodir}:%{?_javadocdir}:/usr/doc:/usr/man:/usr/i
nfo:/usr/X11R6/man
-13: __global_provides_exclude_from %{?_docdir:%{_docdir}}
-13: __global_requires_exclude_from %{?_docdir:%{_docdir}}
-13: _defaultdocdir %{_datadir}/doc
-13: _docdir_fmt %%{NAME}
-13: _javadocdir %{_datadir}/javadoc
-13: _pkgdocdir %{_docdir}/%{name}
...
```

- default format is to include package %{NAME} in docdir
- we can tweak this to suit in our OHPC_macros file

3 components/OHPC_macros

```
@@ -31,7 +31,10 @@
31 31
32 32    %{!?PROJ_DELIM: %global PROJ_DELIM -ohpc}
33 33
   34 + # Update package docs location to include version info (to allow
   35 + # possibility of coinstall of multiple package versions)
34 36    DocDir:    %{OHPC_PUB}/doc/contrib
   37 + %global _docdir_fmt     %{name}-%{version}
35 38
```

Update on Package co-install Gotcha

- now, walk thru co-install of a newer MFEM build goes ok

```
# rpm -q mfem-gnu9-openmpi4-ohpc
mfem-gnu9-openmpi4-ohpc-4.1-8.1.ohpc.2.0.x86_64

# yum list available mfem-gnu9-openmpi4-ohpc.x86_64 --showduplicates
Last metadata expiration check: 0:11:17 ago on Wed 27 Jan 2021 06:27:23 AM CST.
Available Packages
mfem-gnu9-openmpi4-ohpc.x86_64          4.1-8.1.ohpc.2.0           OpenHPC
mfem-gnu9-openmpi4-ohpc.x86_64          4.2-2.2.ohpc.2.1           OpenHPC-update

# yumdownloader mfem-gnu9-openmpi4-ohpc.x86_64
Last metadata expiration check: 0:15:04 ago on Wed 27 Jan 2021 06:27:23 AM CST.
mfem-gnu9-openmpi4-ohpc-4.2-2.2.ohpc.2.1.x86_64.r 4.8 MB/s | 3.5 MB     00:00

# rpm -ivh mfem-gnu9-openmpi4-ohpc-4.2-2.2.ohpc.2.1.x86_64.rpm
Verifying...                          ################################# [100%]
Preparing...                          ################################# [100%]
Updating / installing...
   1:mfem-gnu9-openmpi4-ohpc-4.2-2.2.o################################# [100%]

[ohpc-test@sms005 ~]$ module avail mfem

---------------------- /opt/ohpc/pub/moduledeps/gnu9-openmpi4 ----------------------
   mfem/4.1    mfem/4.2 (D)
```

ARM Compiler Update

- Updated OpenHPC variant of ARM compiler looks to fix install issues reported previously (v20.3.1)
- Available online: https://developer.arm.com/tools-and-software/server-and-hpc/downloads/arm-allinea-studio/openhpc
- On top of basic ohpc requirements, ARM compiler installer assumes availability of python3
- Example minimum install steps:

```
# yum install python3

# ./arm-compiler-for-linux_20.3.1_RHEL-8.sh -a -i /opt/ohpc/pub/arm
--only-install-microarchitectures=generic

# yum install
http://repos.openhpc.community/OpenHPC/2/CentOS_8/aarch64/ohpc-release-
2-1.el8.aarch64.rpm

# yum install arm1-compilers-devel-ohpc
```

ARM Compiler Update

- Loading arm1/compat ohpc module then loads additional ARM provided modules:

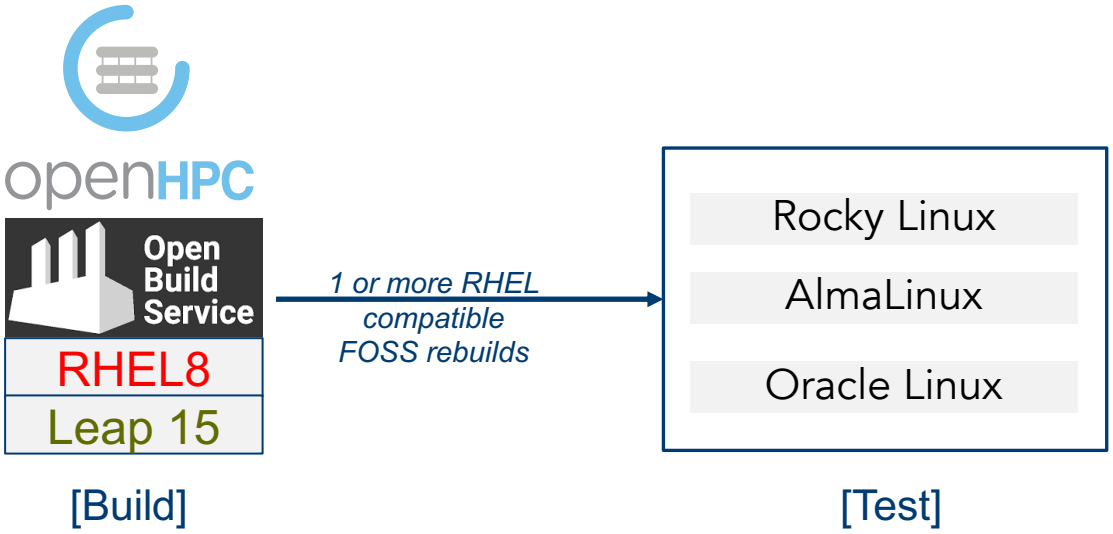
```
$ module swap gnu9 arm1

$ module avail
------------------------ /opt/ohpc/pub/arm/modulefiles ------------------------
   Generic-AArch64/RHEL/8/arm-compiler-for-linux/20.3.1
   Generic-AArch64/RHEL/8/arm-linux-compiler-20.3.1/armpl/20.3.1 (L)
   Generic-AArch64/RHEL/8/arm-linux-compiler/20.3.1               (L)
   Generic-AArch64/RHEL/8/gcc-9.3.0/armpl/20.3.1
   Generic-AArch64/RHEL/8/gcc/9.3.0                               (L)


----------------------- /opt/ohpc/pub/moduledeps/arm1 -----------------------
   metis/5.1.0        openmpi4/4.0.4 (L)    superlu/5.2.1
   mpich/3.3.2-ofi    plasma/2.8.0


------------------------- /opt/ohpc/pub/modulefiles --------------------------
   arm1/compat (L)    libfabric/1.10.1 (L)    os              ucx/1.8.0 (L)
   gnu9/9.3.0         ohpc             (L)    prun/2.0 (L)
```

not us →

us →

CentOS/RHEL Discussion

- Governing board discussion suggested might be useful to assemble some evaluation criteria for which direction we will pivot regarding the CentOS8 announcement
- Also, we didn't really discuss this previously, but thanks to Adrian OpenHPC has RHEL entitlements that could be used for builds...



CentOS/RHEL Discussion

Metric	Option 1	Option 2	Options 1+2
Long-term stability			
Release latency from upstream RHEL			
HPC community adoption			
Drivers for HPC specific needs (file-systems, interconnects, GPUs) Lustre, BeeGFS, OPA, IB			
OHPC collaboration potential			
OHPC test overhead			
Likelihood for problems for OHPC end-users			
Hardware availability for test			
Availability of commercial support			
Support for alternative architectures (and any latency between 1 arch over another)			
Latency for security updates			

OHPC Tutorials for 2021

- PEARC Tutorial submissions due Feb. 9th

- Last year's tutorial was:
  - Intro to OpenHPC and updates
  - Deploying OHPC at AWS using packer and cloud formation
  - Working with the OHPC software stack
  - Few example containerized workloads

Possible topics for 2021

- OHPC as a portable and performant software stack for container native workflows
  - Start with Docker on local system or via https://labs.play-with-docker.com/
  - Move to an AWS tutorial system and run docker containers via podman, charliecloud, and/or singularity
  - Show how the OHPC software repos can be used inside containers
  - Discuss issues related to multi-node MPI jobs, containers, and interconnects
  - Show examples of running containers at TACC and/or PSC?
  - Other ideas?
    - build multi-arch container example
    - run on 2-different ohpc arch clusters

Possible topics cont.

- OHPC as a platform for sustainable HPC and cover OHPC features
  - Feature openSUSE Leap 15.2 (instead of our normal CentOS)
  - Everyone gets another EvenEngine account and we provision an ARM-based elastic cluster for them
  - Do basic benchmarking including network / MPI
  - Install Elastic Fabric Adapter (EFA)
  - Rebuild the OHPC-provided OpenMPI4 package for EFA support
  - Re-do basic benchmarking
  - Install ARM Allinea Studio and show how it integrates with lmod
  - Other ideas?